Exp.no:2 Manual IP Address Configuration

Objective:-

To manually assign a static Internet Protocol (IP) address, subnet mask, Default gateway and DNS server on a host computer to establish network connectivity.

Equipment and Tools:-

- A computer running windows 10 or 11
- An active network connection (Ethernet or wifi)
- Target IP address details from your network administrator.

Procedure:-

Step 1: Open Network Connections:

1. Press windows key + R to open run dialog box.
2. Type ncpa.cpl into the box.
3. Press enter to open the Network Connections window.

Step 2: Access Adapter properties:-

1. Identify your active network adapter
2. Right the active adapter and select properties

Step 3:- Select Internet Protocol Version 4

1. Scroll down the list in properties windows.
2. Click on internet protocol version 4 (TCP/IPv4)
3. Click the properties button below the list

Step 4: Input static IP details:

1. Click the radio button labelled use the following IP address
2. Enter your assigned IP address
3. Enter the subnet mask.
4. Enter the default gateway.
5. Click the radio button labelled use the following DNS server address.
6. Enter the preferred DNS.
7. Click ok to save, the click close on adaptor properties window.

Step 5: Verify the configuration:-

1. Press windows key + X and select terminal or CMD.
2. type ipconfig and press enter
3. Verify that the IPV4 address matches the numbers you entered.
4. Type ping 8.8.8.8 to test your connection to internet.

Observation & Result:-

- The network adapter status will briefly reset & reconnect.
- The ipconfig commands confirms the transition from a dynamic (DHCP) address to the manual static address.
- Successful replies from the ping commands indicates that the static configuration is correct & online.

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